Economics for Engineers

Module 1

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Economic Decision Making

Engineering Costs & Estimation

Economic Decision Making

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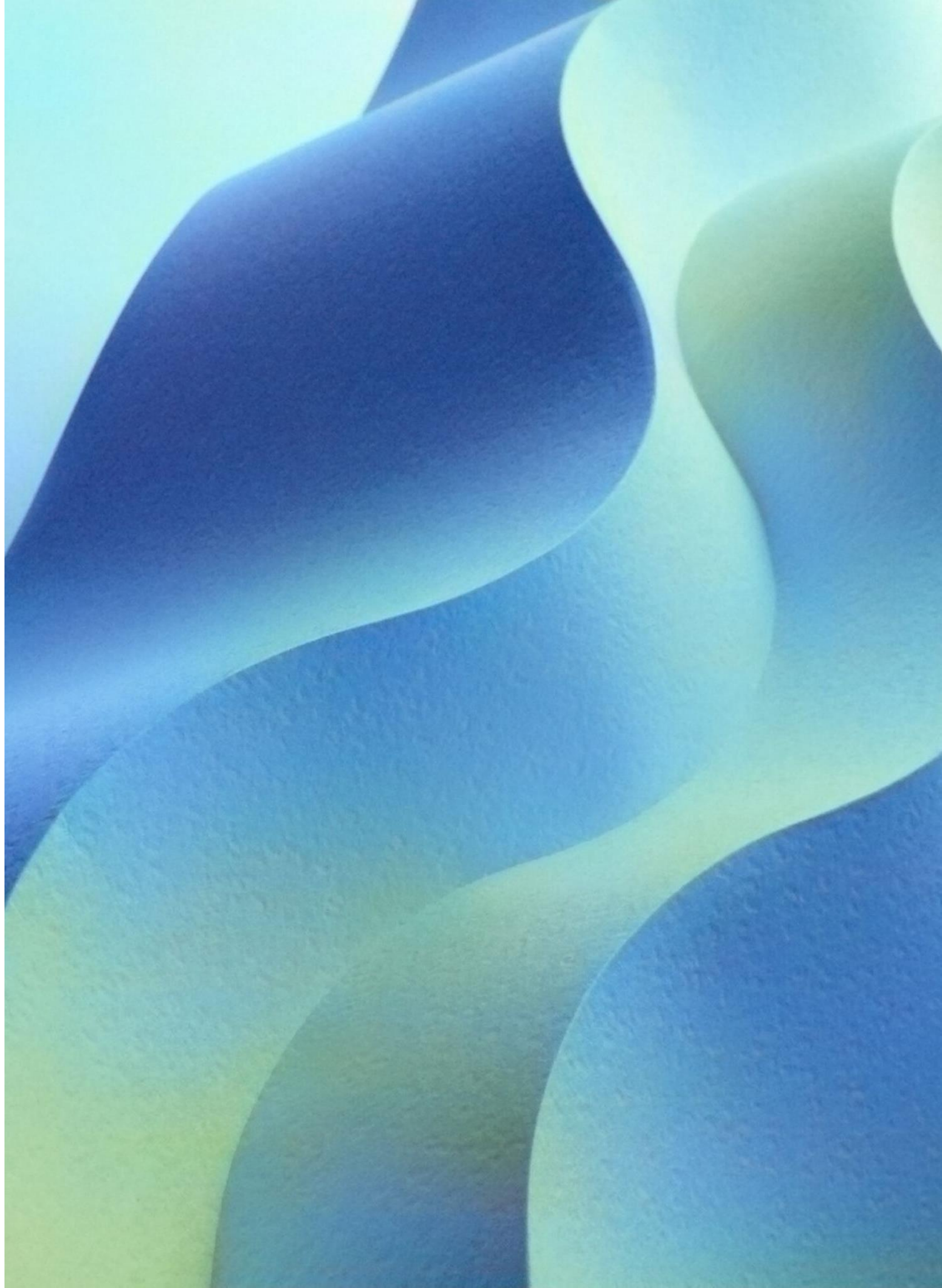

Engineering Costs

- Fixed cost
- Variable cost
- Marginal & Average Costs
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- Recurring And Nonrecurring Costs
- Incremental Costs
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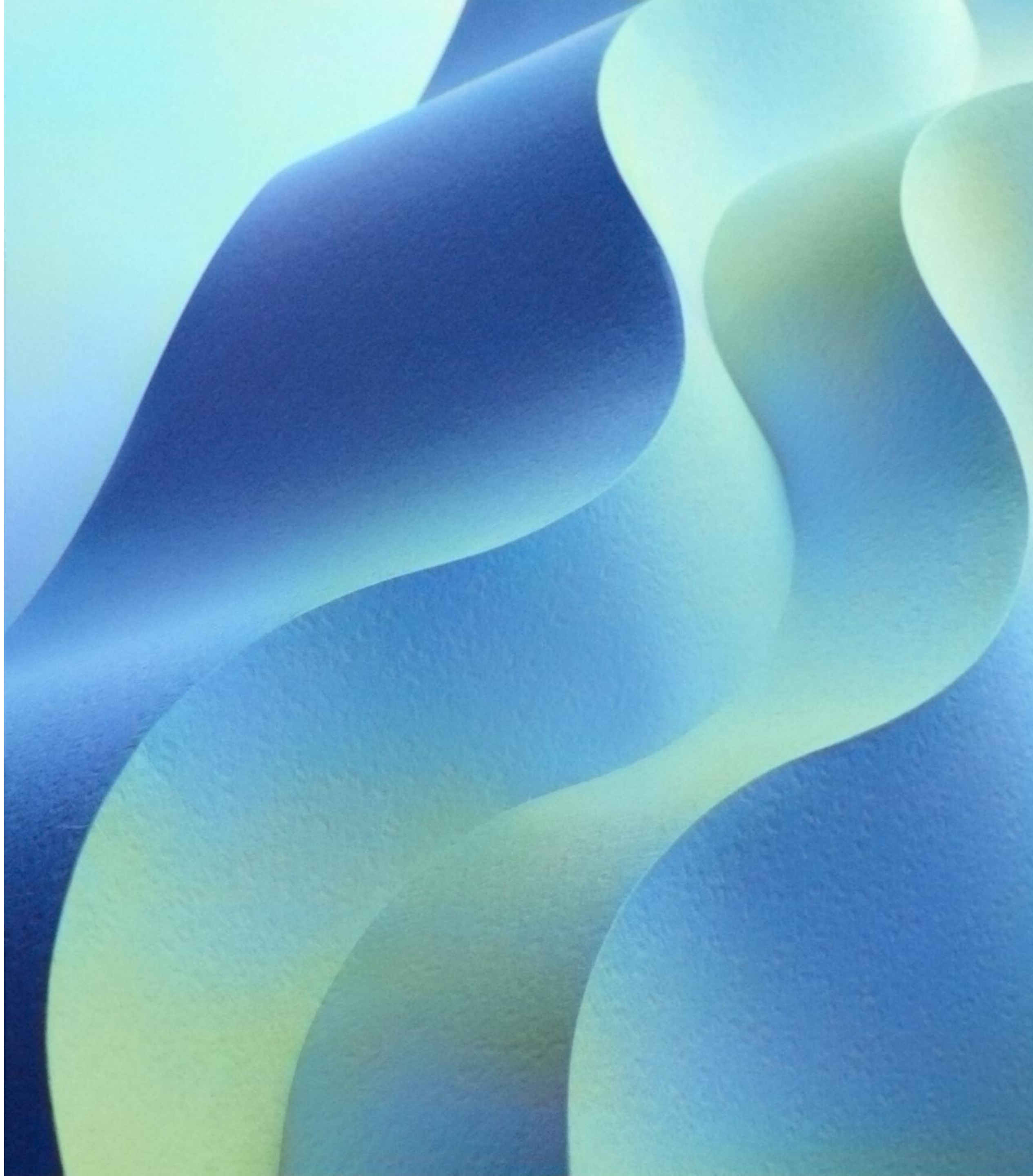


Engineering Estimation

TYPES OF ESTIMATE

ESTIMATING MODELS -

PER UNIT MODEL

SEGMENTING MODEL

COST INDEXES

POWER-SIZING MODEL

IMPROVEMENT & LEARNING CURVE, BENEFITS.

Economic Decision Making

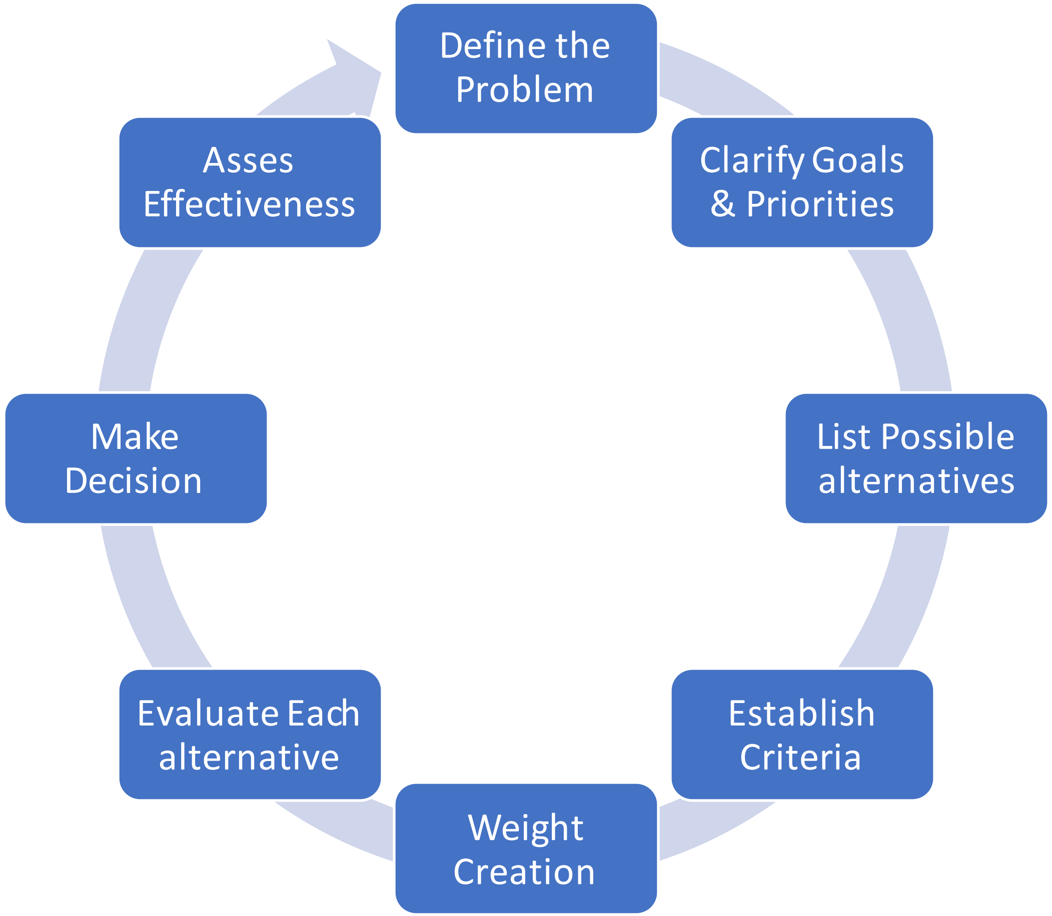
What is Economic?

Economics is the social science that studies the production, distribution, and consumption of goods and services.

What is Decision Making?

the action or process of making important decisions.

Economic Decision Making



City of Dholakpur

Define Problem

What should we develop on the land?

Clarify Goals & Priorities

- Maximize use
- Environmentally Sustainable
- Long term positive impact
- Minimal maintenance Cost
- 'Wow' Factor

Possible Alternatives

PARK & SCHOOL

RETAIL STORES

HOUSING

CORPORATE DEVELOPMENT

List criteria

Cost

Community Need

Environmental impact

Weight Creation

COMMUNITY NEED – 3

ENVIRONMENTAL IMPACT – 2

COST - 1

Evaluation

Alternative	Community need	Environmental impact	cost
Park & School	9	8	4
Retail	6	5	7
Corporate Development	5	5	9
Housing	4	6	8

Weighted Evaluation

Alternative	Community need	Environmental impact	cost
Park & School	9x3 = 27	8x2 = 16	4x1 = 4
Retail	6x3 = 18	5x2 = 10	7x1 = 7
Corporate Development	5x3 = 15	5x2 = 10	9x1 = 9
Housing	4x3 = 12	6x2 = 12	8x1 = 8

Possible alternatives

- Park & School - 47
- Retail stores - 35
- Housing - 32
- Corporate Development - 34

Make Decision

Asses Effectiveness

Engineering Costs

Engineering Costs

- Fixed Cost
- A fixed cost is **a cost that does not change with an increase or decrease in the amount of goods or services produced or sold**.
- Variable Cost
- A variable cost is **a corporate expense that changes in proportion to how much a company produces or sells**

Engineering Costs

Marginal Cost

The marginal cost is **the change in total production cost that comes from making or producing one additional unit**

Average Cost

Average cost is **the cost per unit manufactured in a production run**. It represents the average amount of money spent to produce a product.

Engineering Costs

Sunk cost

A sunk cost is **money that has already been spent and cannot be recovered**.

Opportunity cost

Opportunity cost is **the potential forgone profit from a missed opportunity—the** result of choosing one alternative and forgoing another.

Engineering Costs

Recurring Cost

Recurrent Cost is **the regular expenditure cost that is repeatedly occur for the similar goods or services on a continuing basis**.

Non recurring cost

A nonrecurring cost is **a expense that is considered a one-time or rare expense and is unlikely to happen again**.

Incremental Cost

Incremental cost is **the total cost incurred due to an additional unit of product being produced**

Engineering Costs

Cash Cost

Cash Costs are the costs Which are paid by the business while using cash or cheque, but not credit.

Book Cost

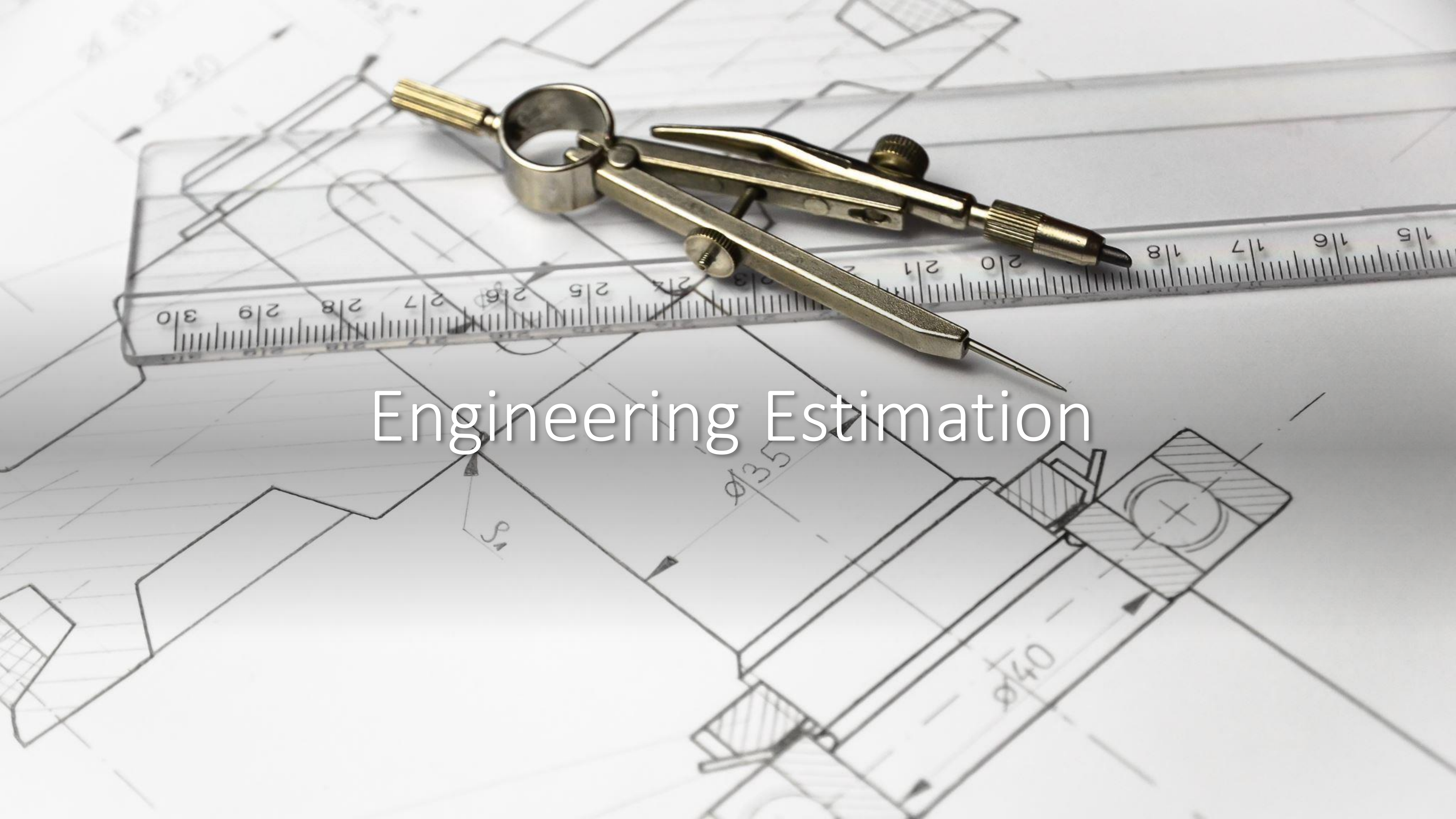
In any business, there are purchases of equipment and other facilities. The total money spent in buying that equipment and other facilities is called Book cost.

Engineering Costs

Life cycle cost

During the life cycle of any product or service, there will be economic costs such as raw material costs, procurement of components and subassemblies, production and assembly of component and subassemblies, usage, maintenance and waste management. The sum of these economic costs is called **life-cycle cost** of that product or service.

Engineering Estimation



Engineering Estimation

In a Manufacturing/service organization, estimating the cost of product/service requires utmost accuracy so that deviation between the actual cost and the estimated cost is minimal.

Engineering Estimation helps us in doing the estimation.

Per Unit Model

The per-unit model of cost estimates the cost of producing and selling one unit.

Segmentation Model

- The segmentation model of cost estimation splits the task of the product/service into different segments. Then the cost of each segment is to be estimated and finally, the cost of the product is obtained by summing the costs of all the segments of product/service.

Cost index Model

- The **cost index describes changes in prices of cost factors relative to the selected base year**.

Power sizing model

- The ‘power sizing model’ of cost estimation is based on **economies of scale.** It is used for estimating the **cost of equipment and other industrial items.**

Learning curve

- Learning Curve is based on the concept that **the more an individual repeats a process or activity, the more adept they become at that activity**. This translates to lower input costs and higher overall output.

Thank you